

IDD-wayfinder-01 - Governed, Schema-Validated, Cost-Audited Construction and Orchestration of Multi-Agent Workflow Pipelines Across Cooperating Advisory and Project-Management Systems, With Optimizer Review and Multi-Tenant Marketplace Composition

Invention Disclosure Document - prepared for patent counsel (preliminary filing and prior-art search).

Status: Draft rev. 3 - internal disclosure, not yet filed or publicly disclosed **Date prepared:** 2026-09-01 (rev. 1); revised 2026-09-01 (rev. 2 - adds governance/constitution layer, schema validation, auditor, optimizer, and marketplace trust model); revised 2026-09-02 (rev. 3 - split from a sibling, CinePipeAi-specific disclosure; this document covers the general-purpose Wayfinder platform technology, assigned to Trusted-Autonomy, Inc.; the CinePipeAi-specific application of this technology is separately disclosed and assigned to CinePipeAi, Inc. in that company's own IP filings - see the note below) **Discloser:** Michael Hunley (michael@hunleys.us) **Assignee:** **Trusted-Autonomy, Inc. Prepared by:** Drafted with AI assistance (Claude) at the discloser's direction, based on discloser's original conception; **this document is not legal advice and has not been reviewed by patent counsel.**

Note on the sibling CinePipeAi disclosure: Trusted-Autonomy, Inc. has conveyed a universal license to CinePipeAi, Inc. covering all Trusted-Autonomy technology, including the Wayfinder platform this document discloses. CinePipeAi's own application of this platform - its "S.A.G.E. x CD" / "S.A.G.E. x PM" system for animated/media production - is separately disclosed, in CinePipeAi-specific terms and assigned to CinePipeAi, Inc., as `IDD-sage-01-cd-pm-workflow-graph.md` in `cinepipe-warden/docs/IP/` (a private repository under the CinePipeAi organization). This document is the general-platform-level disclosure; that one is the CinePipeAi-specific application built on it under license. The two should be read together by counsel and filed, if at all, with a coordinated priority strategy so neither creates a self-prior-art problem for the other - they describe the same underlying mechanism at two different levels of generality, for two different assignees, under one governing license.

Revision note: Rev. 2 incorporated four structural decisions made after rev. 1: (1) the constitution is a non-overridable platform baseline with tenant-specific overlays; (2) the auditor blocks on

correctness/governance violations but is advisory-only on cost; (3) the optimizer is propose-only - it never auto-applies changes; (4) the system is closed to community-authored *Agents*, but a marketplace allows community-authored *Pipes* (graph compositions of first-party Agents plus a user-supplied purpose/summary label). Cross-tenant use of aggregated audit/cost data for platform-wide optimizer improvement is explicitly deferred and will be handled manually, not as an automated system capability, until a separate decision is made. **Rev. 3 (this revision) splits this document from its formerly-shared CinePipeAi framing** - see the note above - and is otherwise unchanged in technical substance from rev. 2.

1. Field of the Invention

Multi-agent AI orchestration; specifically, systems for automatically selecting, composing, and executing directed graphs of specialized AI agents ("Pipes") to accomplish user-stated creative or business goals, where agent and pipeline capabilities are described in a structured, standards-based knowledge graph, and where the resulting execution plan is inserted as tracked work into a separate project-management system via a defined cross-system protocol.

2. Background / Problem Statement

Existing multi-agent orchestration frameworks (e.g., LangGraph, CrewAI, AutoGen, Microsoft Semantic Kernel planners) let a developer hand-wire agents into a graph, or let an LLM pick from a small, hardcoded toolset at runtime. None of the frameworks known to the discloser combine all of the following:

- 1 A **standards-conformant, human- and machine-readable knowledge graph** (built on Google's Open Knowledge Format, OKF) that independently documents each agent's inputs, outputs, purpose(s), and - distinctively - its **pros and cons** (what it is good at, what it is not), at both the individual-agent level and the composed-pipeline level;
- 2 A **dedicated planning/advisory agent** ("Sage CD," a creative-director role) responsible for selecting an existing pipeline or constructing a new one by reasoning over that knowledge graph, rather than by fixed developer-authored routing logic;
- 3 A **separate, cooperating project-management agent** ("Sage PM") that receives the resulting plan as first-class, trackable project tasks, with the pipeline-construction step itself represented as a task whose scope depends on whether a suitable pipeline already exists or must be built;
- 4 A **defined cross-agent communication and task-insertion protocol** between the advisory and project-management systems, scoped by the host platform's existing multi-tenant access-control layer, so that plan authorship and task authorship are cryptographically distinguishable actions;
- 5 **Reusable, learnable project workstream templates** - pre-defined or inferred from observed user behavior - that generalize beyond any single vertical (e.g., not limited to any one licensee's specific production use case) and are intended to be first-class capabilities of the underlying project-management platform (Wayfinder) rather than a vertical-specific bolt-on.

Existing systems either require a developer to hand-author the routing/graph logic, or use an undifferentiated tool list with no structured capability metadata, no cross-system task accountability, and no reusable template learning loop. This creates a gap the invention addresses: a non-technical user can state a creative or business goal, and the system can autonomously determine whether existing capability composition suffices, or new composition must be engineered, and can hold that engineering work accountable as tracked project work rather than an opaque background process.

3. Summary of the Invention

The invention is a system comprising:

- **Agent nodes:** individually deployable units, each defined by an instruction file and represented by a knowledge-graph node describing its required inputs, produced outputs, one or more purposes it serves, and an explicit pros/cons characterization of its suitability.
- **Pipe nodes:** directed graphs of Agent nodes representing an executable workflow, themselves represented at a coarser grain in the same knowledge graph with the same input/output/purpose/pros-cons structure, such that Pipes can be composed from Agents and chained with other Pipes recursively.
- **A graph-construction subsystem** that, given a stated user goal, retrieves candidate Agent/Pipe nodes from the knowledge graph (via embedding-based retrieval and/or a model specifically trained or tuned against the graph) and assembles them into a valid, executable task graph analogous to a task/dependency graph, selecting an existing Pipe where one satisfies the goal and constructing a new one - node by node - where none does.
- **A dual-agent cooperative-planning layer:** an advisory/creative-director agent (Sage CD) responsible for workflow selection and construction, and a project-management agent (Sage PM) responsible for tracking the resulting work as tasks, including the pipeline-construction step itself as a scoped, estimable task distinct from pipeline-execution tasks. The two communicate through a defined protocol scoped by the host platform's access-control primitives (in the reference implementation, OIDC-based identity plus Biscuit-based scoped capability tokens), rather than through shared mutable state.
- **A template layer:** recurring multi-task workstream templates, either pre-authored or inferred from repeated patterns in prior user-directed plans, retrievable and reusable across unrelated projects and tenants (subject to tenant isolation), implemented as a capability of the underlying multi-tenant project-management platform rather than as an application-specific feature.
- **A governance ("constitution") layer:** a versioned, non-overrideable platform-wide baseline rule set, extensible per-tenant with additional (never weaker) rules, evaluated as machine-checkable data - not merely natural-language prompt guidance - against every constructed graph both before it is approved for use and again at execution time.

- **A schema-validation mechanism:** a hard, deterministic check that a constructed graph's declared end-to-end input/output contract satisfies the target goal's required output, performed at construction time as a blocking gate, complemented by a runtime attestation check on actual (not merely declared) outputs each execution, feeding a drift-detection process rather than blocking every individual run.
- **An auditor subsystem:** evaluates every constructed graph along three independent dimensions - schema/output correctness, constitutional compliance, and resource cost - with blocking authority over the first two and advisory (flagging, non-blocking) authority over the third, reflecting that cost tolerance is a business judgment rather than a hard invariant.
- **An optimizer subsystem:** runs only against graphs that have already passed the auditor, evaluates cost/latency/reliability trade-offs (using a tenant-configurable objective weighting) both at construction time and continuously against the library of previously-approved graphs as execution history accumulates, and is strictly propose-only - every optimization is routed through the same human/advisory-agent review gate as new graph construction, never auto-applied.
- **A marketplace composition layer:** distinguishes between first-party Agent nodes (whose inputs, outputs, and core behavioral contract are authored solely by the platform operator and are not community-editable) and community-authored Pipe nodes (which recompose first-party Agents into new graph topologies and carry a community-supplied purpose/summary label). The community-supplied label is treated as an untrusted, human-readable hint for discovery purposes only; it is never trusted as ground truth by the graph-construction or auditor subsystems, which instead rely exclusively on the schema-validation and auditor mechanisms above to determine what a given Pipe actually does before it becomes eligible for automated selection or reuse.

4. Detailed Description

4.1 Knowledge graph representation

Each Agent and each Pipe is represented as a node conforming to (or extending) an open, markdown-plus-YAML-frontmatter knowledge representation format. The reference implementation builds on Google Cloud Platform's Open Knowledge Format (OKF), specifically extending its existing "Attested Computation" concept pattern (which already models parameters, computation reference, executor, and attestation) with additional structured fields for purpose(s) served and an explicit pros/cons list - since the base OKF specification leaves relationship **kind** implicit in prose rather than as a typed, machine-checkable edge property, a point of departure the invention addresses by introducing typed input/output compatibility fields sufficient for automated graph assembly.

4.2 Graph construction

A specialized model or retrieval-and-planning pipeline is trained or configured against the knowledge graph corpus to receive a user (or Sage-CD-derived) goal statement and produce a candidate task graph of Agent nodes - either by selecting a pre-existing Pipe node whose documented purpose and input/output contract satisfies the goal, or, where no such Pipe exists, by composing one from individual Agent nodes such that each edge in the resulting graph satisfies input/output compatibility as documented in the knowledge graph. This is distinguished from generic LLM tool-calling by grounding the construction step in a persistent, versioned, explicitly pros/cons-annotated knowledge graph rather than an in-context tool list, and by producing a durable, reusable, independently inspectable Pipe artifact rather than a one-off tool-call sequence.

4.3 Cross-agent, cross-system planning protocol

Upon receiving a user goal, Sage CD collaborates with Sage PM (optionally using a structured planning methodology) to produce a project plan. One or more resulting tasks are inserted into Sage PM's tracked work; where the plan requires a Pipe that does not yet exist or must be modified, that construction work is itself represented as a distinct, scoped task, with its complexity/scope determined by whether suitable pre-existing Agents/Pipes exist. Task insertion across the CD/PM boundary occurs through a defined API scoped by the host platform's identity and capability-token model, such that plan-authorship and task-insertion are distinguishable, auditable actions rather than shared mutable state between the two agents.

4.4 Reusable workstream templates

Multi-task workstream templates (e.g., "produce an episode of an established animated series with a defined character set") are represented and stored as reusable planning artifacts at the project-management-platform layer, either pre-defined by platform operators or inferred from repeated patterns observed across a tenant's (or, subject to isolation policy, across tenants') prior plans, and are retrievable for reuse on new, related goals without re-deriving the full task breakdown from scratch. This capability is implemented as a general feature of the underlying multi-tenant project-management platform (Wayfinder), not as a feature specific to any single deployed vertical, such that any tenant of the platform benefits from template learning without disclosure of another tenant's templates.

4.5 Governance ("*constitution*") layer

The system maintains a versioned constitution comprising a platform-operator-authored baseline rule set that applies to every tenant without exception, and, per tenant, an optional overlay of additional rules that may only add restrictions, never relax the baseline. Rules are represented as machine-evaluable data (not solely as natural-language guidance to an LLM judge), enabling deterministic, auditable evaluation. A constructed graph is evaluated against the applicable constitution (baseline plus that tenant's overlay) both at construction time, before the graph is approved for use or listed for reuse/marketplace discovery, and again at execution time, so that a graph approved under an earlier constitution version cannot silently continue running after a subsequent tightening amendment. Constitution amendments are themselves versioned, and previously-approved graphs are scheduled for re-audit against a new version rather than grandfathered indefinitely.

4.6 Schema-validated construction and runtime attestation

At construction time, the graph-construction subsystem's candidate assembly is checked against a formal, typed schema derived from the target goal's required output and each node's declared input/output contract; a graph whose end-to-end declared contract does not satisfy the goal's required output is rejected, triggering one automated re-attempt at construction with adjusted candidate retrieval, then escalation to the advisory planning agent (Sage CD) for manual revision, and finally to a human operator if unresolved. Independently, each execution of an approved graph is checked against its actual (runtime) outputs using an attestation mechanism (extending the reference knowledge-representation format's existing attested-computation pattern), and persistent mismatch between declared and actual behavior triggers a drift flag feeding the periodic re-audit process described below, rather than blocking that individual run.

4.7 Auditor

The auditor evaluates every constructed graph, prior to approval, along three dimensions: (a) schema/output correctness as established by 4.6, (b) constitutional compliance as established by 4.5, and (c) resource cost, computed as a composite of compute/token spend and latency normalized to a tenant-configurable budget. The auditor has blocking authority over (a) and (b) - a graph failing either is not approved for use - and advisory-only authority over (c), where a cost overage is flagged to a human reviewer rather than automatically rejected, reflecting that cost tolerance is a tenant business judgment rather than an invariant. In addition to construction-time evaluation, a lightweight version of (a) and (c) runs at every execution, and a full re-audit runs periodically against already-approved, reused graphs to detect drift as underlying Agent nodes or the constitution evolve.

4.8 Optimizer

The optimizer operates only on graphs that have already passed the auditor. It evaluates opportunities to improve cost, latency, and/or reliability - for example, substituting a lower-cost Agent node with an equivalent declared input/output contract, or restructuring a graph for parallel rather than sequential execution - against a tenant-configurable weighting of these objectives. The optimizer runs both at initial construction (proposing an optimized variant of a newly built graph before first use) and continuously in the background against the library of previously-approved graphs as real execution history accumulates. In all cases the optimizer is strictly propose-only: every suggested change is routed through the same review gate as new graph construction and is never applied autonomously, regardless of how low-risk the change may appear.

4.9 Marketplace composition and trust boundary

The system distinguishes two categories of user-contributed content. First-party Agent nodes - their declared inputs, outputs, and core behavioral contract - are authored exclusively by the platform operator; end users and marketplace participants cannot create or modify Agent nodes. Community/marketplace participants may, however, compose existing first-party Agent nodes into new Pipe graph topologies and offer them for reuse or sale via a marketplace, supplying their own human-readable purpose/summary label for discovery. Because the community-supplied label is the only attacker- or error-controllable content in this flow, it is treated by the graph-construction and retrieval subsystems as an untrusted discovery hint only, never as ground truth; a marketplace Pipe's actual behavior, and its eligibility for automated selection or reuse by the graph-construction subsystem, is established solely by whether it passes the schema-validation and auditor mechanisms described above, independent of what its label claims.

5. Novel Features / Points of Distinction

- 1 Dual-granularity knowledge graph (Agent-level and Pipe-level nodes) with explicit, structured pros/cons metadata used as a first-class input to automated selection - not merely descriptive documentation.
- 2 Graph-construction subsystem that treats *node input/output compatibility*, derived from the knowledge graph, as a hard constraint on automated pipeline assembly, rather than relying purely on an LLM's in-context judgment.
- 3 A two-agent cooperative architecture in which a creative/advisory planning agent and a project-management agent are structurally separate systems communicating over a scoped, access-controlled protocol, with pipeline-construction-as-task representable and estimable distinctly from pipeline-execution-as-task.
- 4 Reusable, cross-project, tenant-isolated workstream template learning implemented at the general project-management-platform layer rather than as a vertical-specific feature - i.e., the same underlying capability serves an animation studio's "produce an episode" template and an unrelated tenant's unrelated recurring workstream.
- 5 Extension of an existing open knowledge-representation standard (OKF) to carry structured, typed input/output/purpose/pros-cons fields sufficient for automated graph assembly, rather than either

inventing a wholly proprietary schema or relying on prose-only relationships as the base standard does today.

- 6 A versioned, machine-evaluable governance layer with a non-overridable platform baseline plus tenant-specific overlays, applied as a blocking gate at both graph-construction time and execution time, with amendments triggering scheduled re-audit of previously-approved graphs rather than silent grandfathering.
- 7 A composite auditor with dimension-specific authority - hard-blocking on schema/output correctness and constitutional compliance, advisory-only on resource cost - distinguishing invariant safety/correctness properties from tenant business judgment within a single audit pass.
- 8 A strictly propose-only optimizer, structurally downstream of and gated by the auditor, that evaluates a tenant-configurable cost/latency/reliability objective both at construction time and continuously against accumulated execution history of the approved-graph library, without ever autonomously modifying a live graph.
- 9 A trust-boundary architecture for a multi-tenant agent-graph marketplace that structurally separates platform-authored, non-editable Agent capability nodes from community-composed Pipe topologies, and treats community-supplied descriptive metadata as an untrusted discovery hint that is never consumed as ground truth by the automated construction, auditor, or optimizer subsystems - those instead rely exclusively on the schema-validation and auditor mechanisms to establish actual Pipe behavior. *One licensee's own concrete implementation of this exact trust boundary is separately documented in that licensee's own IP filings (IDD-sage-01, CinePipeAi, Inc., and, on the platform side, IDD-platform-01) - counsel should read all three together when scoping claims, to avoid the same mechanism being claimed independently, without coordination, by two different assignees under one shared license.*

6. Prior Art Landscape (non-exhaustive, informal)

The discloser is aware of, and a formal search should specifically address:

- **Multi-agent orchestration frameworks:** LangGraph, CrewAI, AutoGen, Microsoft Semantic Kernel (planner patterns), and various 2023-2025 patent filings in the LLM-agent-orchestration space by major cloud/AI vendors.
- **Node-graph creative tools:** ComfyUI and similar node-graph-based generative pipelines (prior art for the *visual pipeline* concept, though these are user-authored, not automatically constructed from a knowledge graph).
- **Retrieval-augmented tool selection:** RAG-based tool/function selection systems, which share the "retrieve candidate capabilities" mechanism but are not known to combine it with a typed knowledge graph plus a cooperating, access-controlled project-management system.
- **Workflow-template/BPM systems:** traditional business-process-management template libraries, which share the "reusable recurring workstream" concept but not the AI-agent-composition or knowledge-graph-grounded construction mechanism.

- **AI guardrail/policy-enforcement frameworks:** constitutional-AI and LLM guardrail systems (e.g., rule- or classifier-based content/action filters, policy-as-code engines such as Open Policy Agent applied to AI outputs) share the "govern AI behavior against declared rules" goal but are not known to combine tenant-overlay governance with construction-time *and* execution-time blocking specifically over an automatically assembled multi-agent graph.
- **AI cost-governance/FinOps tooling:** emerging LLM cost-monitoring and budget-alerting products share the advisory cost-flagging concept but are not known to be integrated as one dimension of a unified auditor alongside correctness and governance checks over an agent graph.
- **AI agent/prompt marketplaces:** existing marketplaces for LLM prompts, GPTs, or agent configurations (e.g., major AI platforms' own marketplace/store features) share the "community composes and sells configurations of a platform's building blocks" model, but are not known to combine it with the disclosed trust boundary - where community content is limited to graph topology plus an untrusted label over platform-fixed capability nodes, verified independently by an auditor before becoming eligible for automated reuse.

This is a crowded and fast-moving space, and the guardrail/marketplace categories in particular should be searched carefully - several major AI platform vendors have active product and patent activity here. A professional prior-art search is a prerequisite to any filing decision and may materially narrow or eliminate patentable claims.

7. Advantages / Technical Effects

- Enables non-technical end users to state a creative/business goal in natural language and receive either immediate execution (existing Pipe) or a scoped, estimated engineering task (new Pipe), rather than requiring a developer to hand-wire a workflow.
- Reduces workflow-construction latency and error rate over time, because valid pipelines are captured as durable, reusable, versioned artifacts rather than regenerated per request.
- Provides auditability and accountability for AI-driven work by routing all resulting effort through a conventional, human-reviewable project-management system, consistent with a "human reviews every consequential action" operating posture.
- Generalizes the underlying template-learning and graph-construction capability across unrelated verticals and tenants of a single multi-tenant platform, amortizing the engineering investment across customers rather than confining it to one deployment.
- Provides a deterministic, auditable safety and correctness gate (schema validation plus constitutional compliance) around otherwise-probabilistic LLM-driven graph construction, distinguishing invariant blocking checks from advisory business judgment (cost) within one audit pass.
- Enables continuous, non-disruptive cost/latency/reliability improvement of a growing library of approved graphs via a propose-only optimizer, without the safety risk of autonomous self-modification of live workflows.

- Enables a safe multi-tenant marketplace economy around a fixed, platform-controlled capability set, letting community participants monetize graph-composition expertise without ever expanding the system's trusted-code attack surface.

8. Alternative Embodiments

- The graph-construction subsystem may be implemented as pure retrieval-augmented generation over the knowledge graph with a general-purpose planning model, as a fine-tuned model trained on the graph corpus, or as a hybrid (retrieval for candidate selection, a smaller specialized model for compatibility/validity scoring).
- The knowledge graph representation may be OKF-conformant as described, or may use an alternative open or proprietary graph/markdown representation with equivalent structured fields.
- The advisory and project-management roles (Sage CD / Sage PM) may be collapsed into a single system in a simplified embodiment, though the disclosed advantage of auditable, access-controlled separation would not apply to that embodiment.
- Workstream templates may be scoped globally, per-tenant, or shared selectively across tenants subject to policy.
- The constitution's tenant-overlay mechanism may permit overlays to be authored solely by the tenant, or may itself route tenant-authored overlay proposals through a platform-operator approval step before activation.
- The auditor's dimension-specific authority (blocking vs. advisory) is configurable per dimension and per deployment; a more conservative embodiment could make cost blocking as well, and a more permissive embodiment could make constitutional compliance advisory-only for a fully-trusted internal deployment.
- Cross-tenant use of aggregated audit/cost/execution data for platform-wide optimizer improvement is, in the reference embodiment, explicitly deferred and handled manually rather than automated; an alternative embodiment could formalize this as an opt-in, contractually-governed automated aggregation pipeline.
- The marketplace trust boundary could be relaxed in an alternative embodiment to allow vetted third parties to contribute new Agent nodes (not just Pipe compositions), at the cost of the stronger security posture the reference embodiment achieves by keeping Agent authorship platform-exclusive.

9. Commercial Applications

Primary application: the general-purpose Wayfinder platform, licensable to any tenant or private-label deployment across verticals, since the graph-construction, governance, audit, optimizer, and template-learning mechanisms are designed as general platform capabilities rather than vertical-specific logic. **One reference deployment, under a universal license from Trusted-Autonomy, Inc.: CinePipeAi, Inc.'s "S.A.G.E. x CD" / "S.A.G.E. x PM" private-labeled deployment**, for animated/media production workflows (e.g., recurring "produce an episode" templates for an established character set, and a marketplace of community-composed production Pipes built from a fixed set of platform-authored media-production Agents) - separately disclosed in CinePipeAi-specific terms as IDD-sage-01 in that company's own IP filings. The governance/auditor layer specifically also has standalone commercial relevance as a trust and safety feature for any multi-tenant AI orchestration offering, independent of any single licensee's vertical.

10. Public Disclosure Status

No public disclosure, publication, demonstration, or sale of this system has occurred as of the date of this document, to the discloser's knowledge. This status is time-sensitive and should be reconfirmed at the time of any filing decision, since public disclosure can start statutory bars to patentability in some jurisdictions.

11. Patentability and Protection-Strategy Evaluation

This section is a technical, non-legal assessment intended to inform a conversation with qualified patent counsel. It is not legal advice and should not be relied upon as a substitute for one.

Patent considerations

- **Novelty/non-obviousness risk remains real for the individual mechanisms, but the composite system is a materially stronger claim than rev. 1.** Multi-agent orchestration, RAG-based tool selection, node-graph pipelines, workflow-template libraries, AI guardrail/policy engines, LLM cost-monitoring tools, and AI configuration marketplaces are all independently well-established in the prior art (see Section 6) - no single mechanism here is likely novel in isolation. What appears more distinctive is the **specific combination**: a single system in which (a) graph construction is hard-gated by typed schema validation against a stated goal, (b) the same construction is independently hard-gated by a versioned, tenant-overlaid, machine-evaluable governance layer enforced at both construction and execution time, (c) a composite auditor applies different authority levels (blocking vs. advisory) per dimension within one pass, (d) a structurally downstream, strictly propose-only optimizer improves the approved-graph library over time without ever autonomously modifying a live graph, and (e) a marketplace lets third parties compose - but never author - the underlying capability nodes, with their descriptive metadata treated as untrusted. This composite claim (Section 5, items 6-9 in particular, combined with items 1-3) is narrower than any individual mechanism and correspondingly harder for a competitor to design around without dropping one of the elements.

- **Patents require eventual public disclosure** (publication ~18 months after filing in most jurisdictions, or at grant), which cuts against maintaining the more implementation-specific advantages (exact KG schema extensions, exact governance rule representation, exact auditor/optimizer scoring logic) as confidential.
- **Enforcement difficulty is reduced relative to rev. 1's assessment, but not eliminated.** Some elements remain server-side and unobservable (the retrieval/fine-tuning approach, the optimizer's scoring logic). But others may be at least partially observable or inferable from product behavior by a sophisticated competitor or customer - for example, a "construction rejected: does not satisfy your organization's policy" error message, a visible marketplace Pipe-approval status, or a visible cost-flag UI would each hint at the existence (though not the internal mechanism) of the governance, auditor, and cost-flagging components. This modestly increases the practical enforcement value of patent protection relative to rev. 1's single-mechanism assessment, since observable *behavior* patterns (a specific sequence of gates and authority levels) may support an infringement argument even where the underlying algorithm is hidden.
- **Filing cost and timeline:** a provisional filing is relatively low-cost and preserves priority date optionality for 12 months without immediate publication - a reasonable low-risk step to preserve the option, especially now that the composite claim is more specific and defensible than the rev. 1 assessment suggested.

Trade secret considerations

- The specific KG schema extensions, the graph-construction model/retrieval configuration, the exact governance rule representation and evaluation engine, the auditor's scoring thresholds and weighting logic, the optimizer's objective functions, and - most importantly - any tenant-specific learned workstream templates and tenant-specific constitution overlays are all naturally difficult for a competitor to reverse-engineer from outside observation, which continues to favor trade-secret protection for these specific implementation details regardless of the patent decision below.
- Trade secret protection is indefinite (no 20-year cap) as long as reasonable secrecy measures are maintained, requires no disclosure, and imposes no immediate filing cost - but offers no protection against independent invention or lawful reverse engineering, and provides materially weaker leverage in fundraising/deterrence conversations than an issued patent.
- Given the KG- and template-based data itself may embed customer-specific (e.g., a licensee's own) workflow structure, and given tenant-specific constitution overlays are themselves potentially sensitive (they may reveal a customer's content-safety or business-risk posture), trade secret status has a direct data-governance implication: contractual confidentiality terms with private-label licensees (including the universal license to CinePipeAi, Inc.) should explicitly cover both categories, independent of the IP-protection decision.

Recommendation

Pursue trade secret protection for implementation specifics as before, and treat a narrow provisional patent filing on the composite system (the combination described in "Novelty/non-obviousness" above, corresponding to Section 5 items 1-3 and 6-9 together) as a more clearly justified near-term action than rev. 1 suggested, given the combination is more specific, more behaviorally observable, and sits across a less-crowded intersection of prior art than any single mechanism alone. This is still a recommendation to evaluate with counsel and a professional prior-art search, not a decision to file. Immediate practical steps that don't require a filing decision, in addition to rev. 1's: mark the governance rule set, auditor scoring logic, and optimizer objective functions as confidential on the same basis as the KG schema and template data; ensure the CinePipeAi license agreement explicitly covers CinePipeAi's own tenant-specific constitution overlays and learned templates as confidential CinePipeAi data, not Trusted-Autonomy data; and coordinate claim scope and priority strategy with counsel across this document, IDD-sage-01 (CinePipeAi, Inc.), and IDD-platform-01 (CinePipeAi, Inc.) before any of the three is filed, given all three touch the same underlying trust-boundary mechanism (Section 5, item 9).

12. Legal Disclaimer

This document was prepared with AI assistance at the discloser's direction and reflects a technical description and informal analysis only. It does not constitute legal advice, does not constitute a formal patentability opinion, and has not been reviewed by a licensed patent attorney. Before making any filing decision, disclosing this invention publicly, or relying on any statement in Section 11, consult qualified patent counsel and, if pursuing a filing, commission a professional prior-art search.